

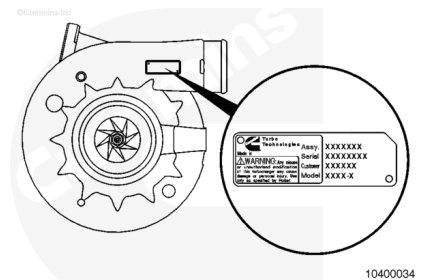
Trainings ▾

Install

All Applications Except Marine and Rail

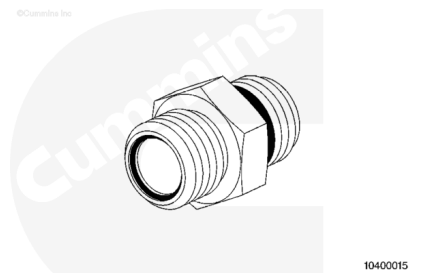
Cummins Turbo Technologies™ turbocharger(s) are used on the QSK19 engine. See the data tag on the turbocharger to determine the model.

The name is also cast on the housing of a turbocharger.



All turbocharger fittings are the flat face o-ring type fittings. Be sure the o-ring is in place before attaching the hose(s) to the fittings.

The Cummins Turbo Technologies™ turbocharger uses a 9/16-18 UNF, straight thread, o-ring type oil supply fitting.



Apply an anti-seize compound to the mounting capscrews.

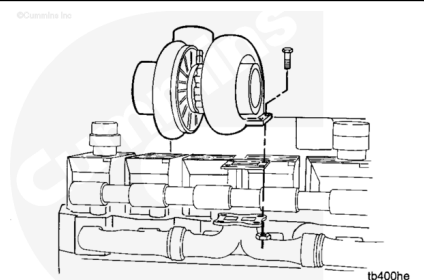
Install the gasket.

Note : The word **OUT** must be positioned toward the turbocharger or the raised gasket bead.

Install the turbocharger, washers, and nuts.

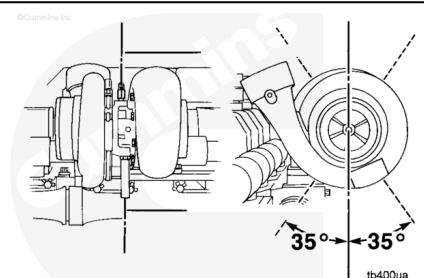
Tighten the capscrews.

Torque Value: 40 n•m [30 ft•lb]



Position the turbocharger drain tube. The drain tube **must** be within 35 degrees of vertical. Turn the bearing housing to align the tube, if necessary.

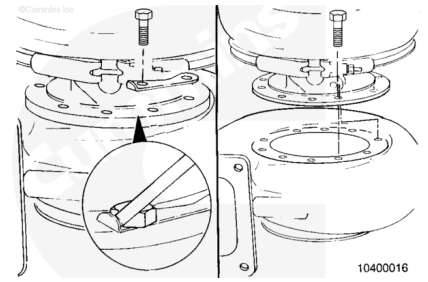
The turbocharger uses capscrews to attach the bearing housing to the turbine housing.



To adjust the bearing housing, bend the lockplates from the capscrew heads.

Remove the capscrews.

Rotate the bearing housing. Align the drain tube and the capscrew holes.

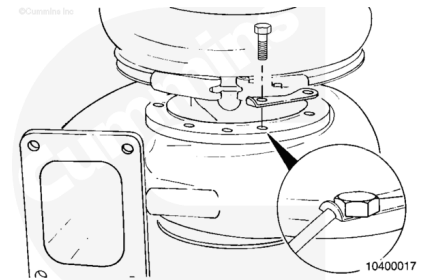


Install the lockplates and capscrews.

Tighten the capscrews.

Torque Value: 20 n·m [177 in-lb]

Bend the lockplate tabs over the capscrews.



⚠ CAUTION ⚠

These parts are a safety-related item. Be sure the parts are clamped correctly to prevent fretting.

Install the oil drain tube and gasket. Refer to Procedure 010-045 in Section (</qs3/pubsys2/xml/en/procedures/20/20-010-045.html>)

Tighten the capscrews.

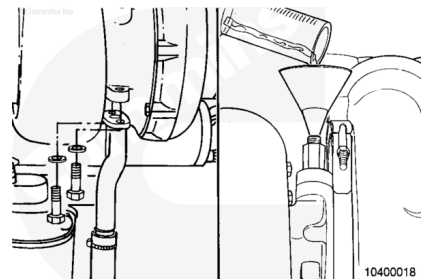
Torque Value: 45 n·m [33 ft-lb]

Tighten the drain hose swivel to the oil pan adapter.

Torque Value: 95 n·m [70 ft-lb]

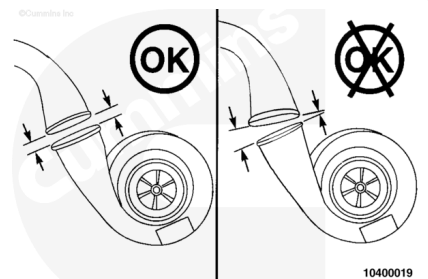
Pour clean engine oil into the turbocharger oil supply fitting.

Turbocharger Oil Supply: 50 to 60 cc [2.0 to 3.0 oz.]



Check the air crossover-to-turbocharger alignment.

- Install the air crossover on the aftercooler.
- Look at the alignment between the crossover and the turbocharger compressor outlet.
- Remove the crossover and proceed with the assembly if the alignment is correct.
- Proceed to the next step to adjust the compressor housing if the alignment is **not** correct.



To adjust the turbocharger compressor housing alignment:

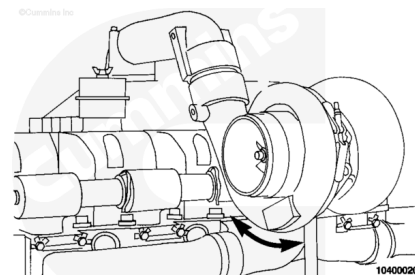
Loosen the v-band clamp.

Turn the compressor housing to the correct alignment.

Tighten the clamp.

Torque Value: 8 n•m [71 in-lb]

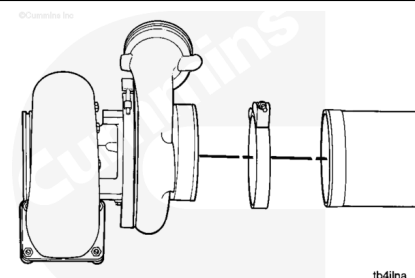
Tap the clamp with a mallet and tighten the clamp again.



Connect the inlet piping to the turbocharger.

Tighten the clamps.

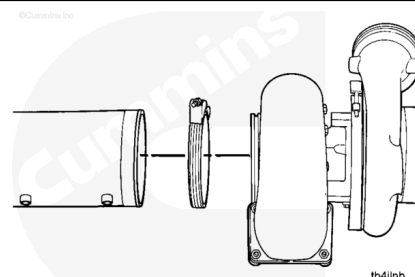
Torque Value: 8 n•m [71 in-lb]



Connect the exhaust piping to the turbocharger.

Tighten the v-band clamp.

Torque Value: 8 n•m [71 in-lb]



Marine Applications

Apply an anti-seize compound to the mounting capscrews.

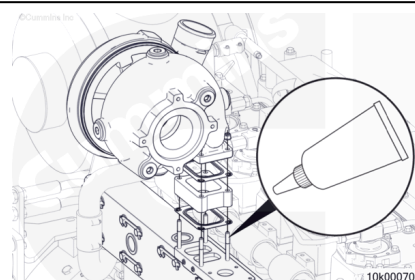
Install the gasket(s).

Note : The word **OUT** must be positioned toward the turbocharger or the raised gasket bead.

Install the turbocharger, washers, and nuts.

Tighten capscrews in a criss cross pattern.

Torque Value: 40 n•m [30 ft-lb]

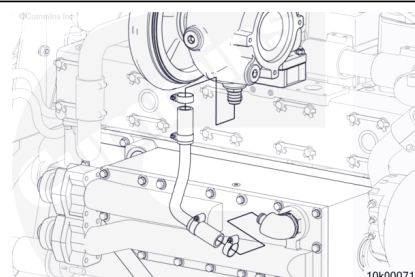


For engines with auxiliary cooler:

Install turbocharger coolant supply line.

Torque Value:

Hose Clamp Torque: 5 n•m [45 in-lb]

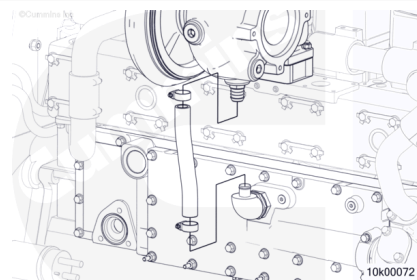


For engines without auxiliary cooler:

Install turbocharger coolant supply line.

Torque Value:

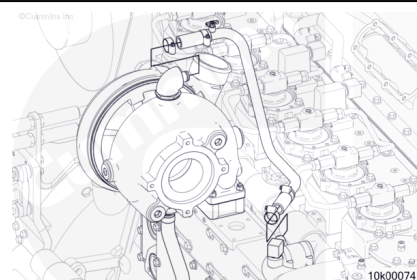
Hose Clamp Torque: 5 n•m [45 in-lb]



Install turbocharger coolant drain line.

Torque Value:

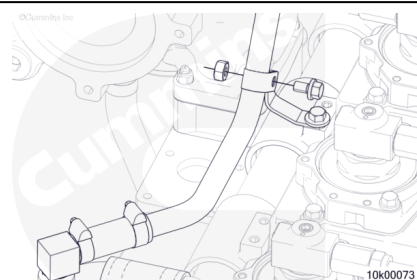
Hose Clamp Torque: 5 n•m [45 in-lb]



Install P-clip

Torque Value:

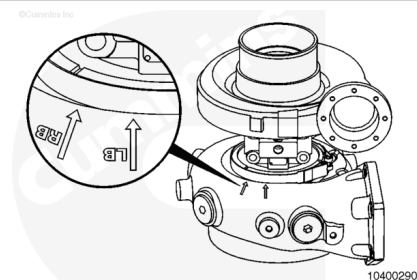
Hose Clamp Torque: 40 n•m [30 ft-lb]



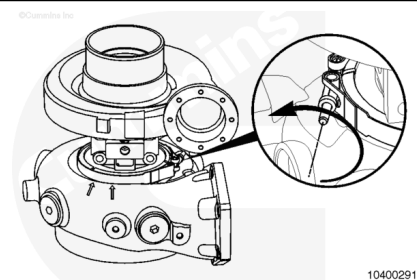
Note : All views showing the clock marks have been simplified for clarity.

Note : The turbocharger for the QSK19 Marine application comes clocked from the factory for the QSK19.

Verify the correct clocking of the turbocharger.

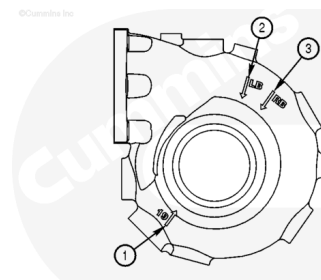


If the turbocharger requires clocking, set the turbocharger on a bench with the turbine housing against the bench and loosen the v-band clamp.

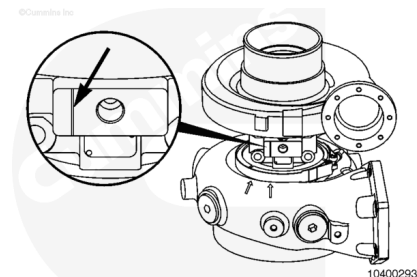


The turbocharger can be used in different applications and has three clock marks on the turbine housing.

1. 19 casting line
2. Left bank
3. Right bank



The bearing housing has a machined surface on the top, creating an alignment mark.



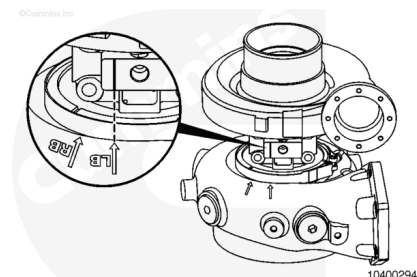
Rotate the bearing housing, aligning the mark with the 19 casting line on the turbine housing.

Apply anti-seize compound to the v-band threads.

Tighten the v-band clamp.

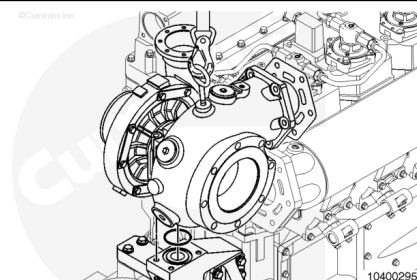
Torque Value:

1. 20 n•m [177 in-lb]
2. Loosen 2 to 3 turns
3. 18 to 20 n•m [159 to 177 in-lb]



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



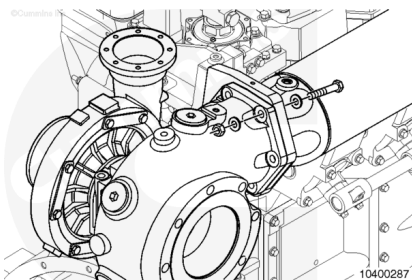
Lubricate and install a new o-ring into the machined groove on the turbocharger center support/drain assembly.

Install the turbocharger and exhaust manifold gasket.

Install the six capscrews, washers, and nuts. Tighten the nuts.

Torque Value:

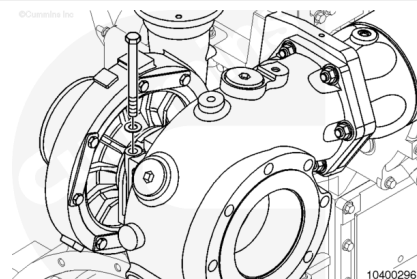
Turbocharger Capscrews 136 n•m [100 ft-lb]



Install the turbocharger to the center section mounting capscrews and washers and tighten.

Torque Value:

Turbocharger to Center Section Mounting Capscrews 115 n•m
[85 ft-lb]

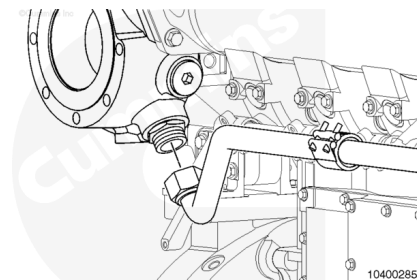


Install the turbocharger turbine housing coolant return line at the turbocharger.

Torque Value: 43 to 47 n•m [32 to 35 ft-lb]

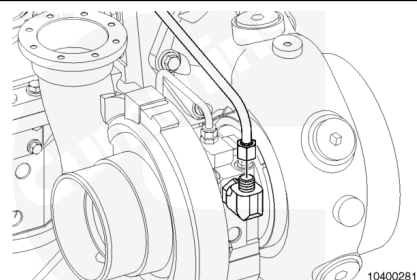
Install a new gasket and capscrews on the coolant bypass tube connection.

Torque Value: 40 n•m [30 ft-lb]



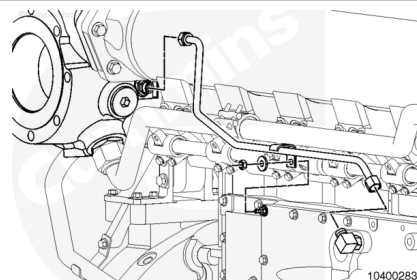
Install the turbocharger bearing housing coolant supply line.

Torque Value: 40 n•m [30 ft-lb]



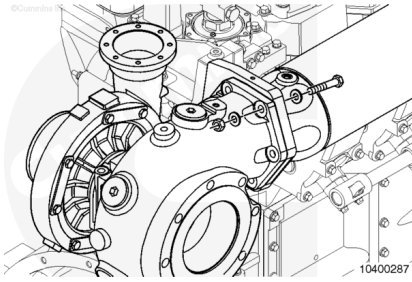
Install the turbocharger bearing housing coolant return line.

Torque Value: 43 to 47 n•m [32 to 35 ft-lb]



Install the three capscrews to the support bracket to compressor housing.

Torque Value: 45 n•m [33 ft-lb]



Last Modified: 14-Mar-2018
